

Rotation-Invariant Composite Kernel Findings

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Questions

- Can we make the composed kernel rotation invariant using right-angle rotations?
- What is test-set performance after enforcing rotation invariance?

Method

A rotation-invariant composite kernel was constructed from the best single composed kernel:

$$K_{\text{RI}} = \frac{1}{4} \sum_{k=0}^3 \text{rot90}(K_{\text{comp}}, k)$$

where $k \in \{0, 1, 2, 3\}$ corresponds to $0^\circ, 90^\circ, 180^\circ, 270^\circ$.

- trains a single-feature classifier on train patches using K_{RI} ,
- evaluates on the original test split,
- evaluates again on $90^\circ, 180^\circ$, and 270° rotated versions of the same test patches.

Rotation-Invariant Kernel Visualization

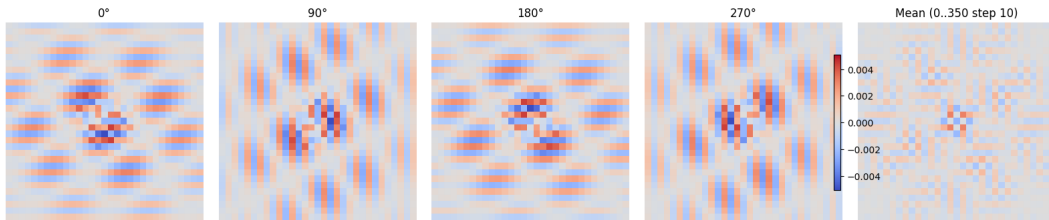


Figure 1: Rotation-invariant composite kernel K_{RI} used for the test-set evaluations.

Data and Model Setup

- Train split: $X_{\text{in}} = (2800, 128, 128)$, $X_{\text{out}} = (3000, 128, 128)$
- Test split: $X_{\text{in}} = (1040, 128, 128)$, $X_{\text{out}} = (830, 128, 128)$
- Test sample count: $N = 1870$
- Classifier type from config: `logistic`

Results

Evaluation	Test AUC	Test ACC
Original invariant kernel (test 0°)	0.890774	0.809091
Invariant kernel, test rotated 90°	0.890774	0.809091
Invariant kernel, test rotated 180°	0.890774	0.809091
Invariant kernel, test rotated 270°	0.890773	0.809091

Rotation deltas (rotated – original invariant kernel):

- 90° : $\Delta\text{AUC} = +0.000000$, $\Delta\text{ACC} = +0.000000$
- 180° : $\Delta\text{AUC} = +0.000000$, $\Delta\text{ACC} = +0.000000$
- 270° : $\Delta\text{AUC} = -0.000001$, $\Delta\text{ACC} = +0.000000$

Findings

- The averaging construction produced a kernel that is effectively right-angle rotation invariant in evaluation.
- Test accuracy is exactly unchanged across 0° , 90° , 180° , and 270° .
- Test AUC is also stable, with only a negligible numerical difference at 270° (-10^{-6}).

Work in Progress

- **Smarter kernel sampling:** replacing purely random parameter sampling with low-discrepancy strategies such as QMC or Latin Hypercube Sampling to improve coverage of each family parameter space.
- **Per-family normalization:** applying normalization rules tailored to each kernel family before composition and response extraction, to better balance family-specific amplitude and scale differences.

Conclusion

The rotation-invariant composite kernel achieved stable test performance across all right-angle rotations in this run. The invariance objective was met on the test set.